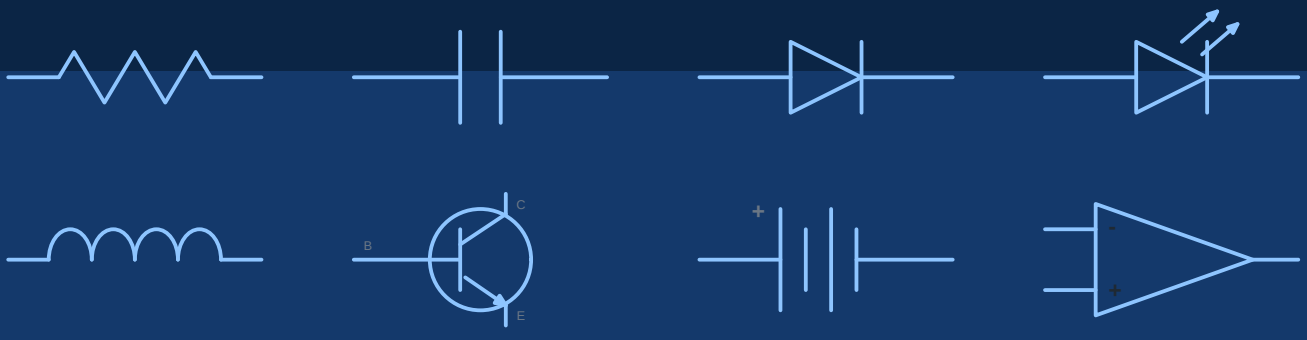
Electronics Components Handbook

Symbols, Jobs & Real-World Uses

54 components | 8 categories | November 2026



Contents

Components are grouped by category. Each entry shows the circuit symbol, a plain-English job, and a real-world example.

Power 3

Control 5

Protection 6

Passive 7

Sensor 8

Actuator 9

Connection 10

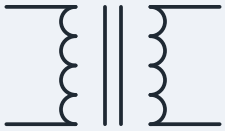
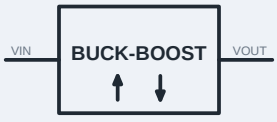
IC 11

Quick Reference Cheat Sheet 12

About the symbols. Resistors, inductors and similar parts use the common ANSI-style drawings; the fuse, varistor and thermistor use the IEC rectangle style. Complex parts (sensors, MCU, SoC, ASIC, FPGA, memory) use labelled block symbols with example pin names. Real parts differ in pin count and naming, so always check the datasheet. Battery and Ground are added to the Power section as everyday companion symbols.

Power

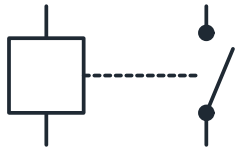
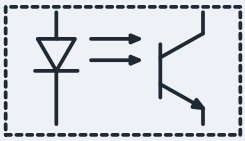
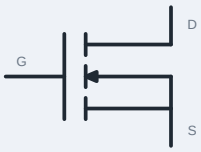
Parts that take electricity from the wall or a battery and reshape it into the voltage and type (AC or DC) a circuit needs.

Symbol	Name & appearance	Job	Real-world example
	AC Mains *Looks like: A wall socket plus a plug on a 2- or 3-core cable (live, neutral, earth).*	Supplies 240 V, 60 Hz alternating current to power the whole circuit.	The wall outlet that powers a TV, a fridge or a phone charger.
	**Transformer** *Looks like: A heavy block of laminated iron wrapped in copper wire, or a squat plastic-and-metal cube with 4 or more pins. Mains types often hum.*	Steps AC voltage up or down using two coils that share a magnetic core.	The block inside an old phone charger that drops 240 V down to about 12 V.
	Rectifier (Diode Bridge) *Looks like: A small square or flat plastic block with four legs marked ~ ~ + -, or four separate diodes wired in a diamond.*	Converts AC into pulsating DC by letting only one direction of current through.	The first stage of a laptop power brick, right after the transformer.
	Filter Capacitor *Looks like: A tall aluminium can with a stripe marking the negative leg and values like 2200 uF 25 V printed on the side.*	Smooths bumpy, pulsating DC into steady DC by storing and releasing charge.	The big capacitor in a phone charger or amplifier that removes mains hum.
	Voltage Regulator (LDO) *Looks like: A small three-legged black package (TO-220 or TO-92), or a tiny surface-mount chip labelled like 7805 or AMS1117.*	Holds a fixed, steady output voltage even when the input or the load changes.	A 3.3 V regulator giving clean power to the chip on a Wi-Fi board.
	Buck Converter *Looks like: A small module or board with a coil (inductor), a controller chip and a few capacitors.*	Steps a DC voltage down efficiently by switching rapidly through an inductor.	A USB car charger turning the 12 V car battery into 5 V.
	Boost Converter *Looks like: An inductor, a diode, a switching chip and capacitors, often on a tiny board.*	Steps a DC voltage up efficiently by storing energy in an inductor and releasing it at a higher voltage.	A power bank lifting a 3.7 V lithium cell to 5 V for USB.
	**Buck-Boost Converter** *Looks like: A module with an inductor, switching chip and capacitors, similar to buck and boost boards.*	Steps DC up or down as needed so the output stays steady while the input wanders above or below it.	A battery gadget holding 3.3 V as its cell drains from 4.2 V to 2.8 V.

Symbol	Name & appearance	Job	Real-world example
	Inverter *Looks like: A metal box with cooling vents, a DC input (clips or car plug) and an AC socket on the front.*	Converts DC from a battery or solar panel back into AC.	Running a laptop or a fan from a car battery or solar system during a power cut.
	Ground (0 V reference) *Looks like: Not a part, but a connection point: the common return path or 0 V reference for the circuit.*	Gives every part in the circuit the same zero-volt reference to measure from.	The black wire or metal chassis that all signals return to.
	Battery *Looks like: A cylindrical (AA), rectangular (9 V) or flat pouch (lithium polymer) cell with + and - terminals.*	Stores chemical energy and supplies steady DC electricity.	A remote control, a phone or a torch.

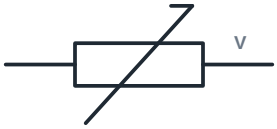
Control

Parts that decide when and where current flows: switching, steering and isolating signals.

Symbol	Name & appearance	Job	Real-world example
	Switch *Looks like: A toggle, rocker or slide lever that stays in whatever position you leave it.*	Opens or closes the circuit, turning current fully on or off.	The switch on a desk lamp or the wall light switch.
	Push Button *Looks like: A small square tactile button with 4 legs, or a round panel button that springs back when released.*	Connects the circuit only while you hold it down (momentary input).	A doorbell button, a keyboard key or a router's reset button.
	**Relay** *Looks like: A small blue or black cube with about five pins; you hear a click when it switches.*	A small current in a coil moves a mechanical contact that switches a separate, bigger load.	A microcontroller turning on a water pump or air-conditioner compressor.
	**Optocoupler** *Looks like: A small black 4-pin (or 6-pin) chip, like a tiny DIP IC.*	Passes a signal between two circuits using light, so they stay electrically isolated.	Keeping a microcontroller safely isolated from the mains side of a power supply.
	Transistor (BJT) *Looks like: A black half-cylinder (TO-92) with three legs, or a larger metal-tab version for more power.*	A small current into the base controls a much larger current through the collector, so it switches or amplifies.	An amplifier stage in a radio, or a transistor switching an LED or relay.
	**MOSFET** *Looks like: Similar to a BJT but often TO-220 with a metal tab, or a flat surface-mount package; three legs: gate, drain, source.*	A voltage at the gate turns a high-current path on or off with very little loss.	Motor drivers in a drone, and switching power supplies in PCs.
	Thyristor (SCR) *Looks like: A TO-220 style package with three legs: anode, cathode, gate.*	Once a pulse at the gate turns it on, it stays on until the current drops to zero, so it latches.	A light dimmer or the speed control in a power tool.

Protection

Parts that sacrifice themselves or clamp dangerous levels so the rest of the circuit survives.

Symbol	Name & appearance	Job	Real-world example
	Fuse *Looks like: A small glass tube with metal end caps and a thin wire inside, or a coloured plastic blade fuse.*	Contains a thin wire that melts and breaks the circuit when current gets too high.	The fuse in a plug top, or in a car's fuse box.
	Circuit Breaker *Looks like: A switch-like lever on a rail-mounted module inside the distribution board.*	Trips open when current is too high and can be reset by hand instead of replaced.	The breaker in a home fuse board that trips when a kettle shorts.
	**Varistor (MOV)** *Looks like: A blue or orange disc with two leads, resembling a ceramic capacitor.*	Drops its resistance when voltage is too high, soaking up voltage spikes (surge protection).	Inside a surge-protected power strip, protecting devices from lightning spikes.
	Zener Diode *Looks like: A small glass-body diode with a dark band at one end and a value printed like 5V1.*	Blocks current like a normal diode, but allows reverse current once a precise voltage is reached.	Clamping a microcontroller input to 5.1 V, or a simple voltage reference.

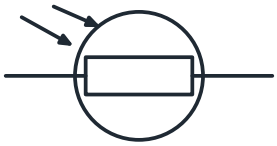
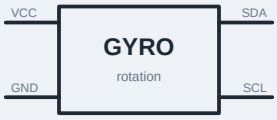
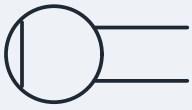
Passive

Parts that shape current, voltage and timing without amplifying anything. (Diodes and crystals are grouped here as basic discrete parts.)

Symbol	Name & appearance	Job	Real-world example
	Resistor *Looks like: A small cylinder with coloured bands (through-hole), or a tiny black rectangle with a number (surface-mount).*	Limits current and drops voltage.	A 330 ohm resistor in series with an LED so it does not burn out.
	Potentiometer *Looks like: A round knob with three pins, or a small blue trimmer with a screw slot.*	A resistor whose value you change by turning a knob or screw.	The volume knob on a radio.
	Capacitor *Looks like: A yellow or brown ceramic disc, a small rectangular film block, or a tiny surface-mount chip.*	Stores charge, filters signals, and blocks DC while passing AC.	A small decoupling capacitor sitting beside nearly every chip.
	Inductor (Coil) *Looks like: A coil of copper wire on a core, a ring-shaped toroid, or a small square surface-mount block.*	Stores energy in a magnetic field and resists sudden changes in current.	The coil inside a buck converter, or in a radio tuning circuit.
	Diode *Looks like: A small glass or black cylinder with a stripe on the cathode end.*	Lets current flow in one direction only.	Reverse-polarity protection in a battery-powered toy.
	Crystal Oscillator *Looks like: A small silver metal can (HC-49) or a tiny rectangular surface-mount part, stamped with a frequency like 16.000.*	Vibrates at one very exact frequency to give a precise timing or clock signal.	The 16 MHz crystal on an Arduino, or the 32.768 kHz crystal in a digital watch.

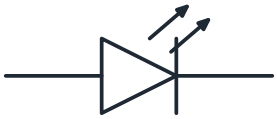
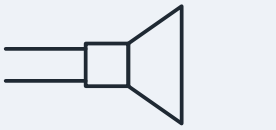
Sensor

Parts that turn the real world (light, heat, motion, sound, pressure) into electrical signals.

Symbol	Name & appearance	Job	Real-world example
	**LDR (Photoresistor)** *Looks like: A small disc with a zig-zag track of dark material under a clear window, with two leads.*	Its resistance goes down as more light falls on it.	A streetlight that switches on at dusk.
	Thermistor *Looks like: A small bead or disc with two leads, often glass- or epoxy-coated.*	Its resistance changes predictably with temperature.	The temperature probe in a coffee machine or a battery pack.
	Accelerometer *Looks like: A tiny square chip, a few millimetres wide, often on a small breakout board.*	Senses acceleration, which lets it detect motion, shock and tilt.	A phone rotating its screen, or a fitness band counting steps.
	**Gyroscope** *Looks like: A tiny chip, often combined with an accelerometer on one board (an IMU such as the MPU6050).*	Senses how fast something is rotating.	A drone staying level, or a camera gimbal keeping a shot steady.
	Humidity Sensor *Looks like: A small blue or white plastic box with a perforated grille (like a DHT22), or a tiny surface-mount chip.*	Measures how much moisture is in the air.	A weather station or a smart thermostat.
	Pressure Sensor *Looks like: A small metal can or tiny silver chip with a pinhole port, or a thin flat force pad.*	Measures pressure or force and outputs a matching electrical signal.	A phone barometer for altitude, or a tyre-pressure monitor.
	Image Sensor (CMOS) *Looks like: A small camera module with a lens, or a bare silicon chip under glass with a grid of tiny light-sensitive pixels.*	Turns the light of a scene into a grid of digital pixels.	The camera in a phone, a webcam or a video doorbell.
	**Microphone** *Looks like: A small metal can with a mesh-covered hole (electret), or a tiny surface-mount chip with a pinhole.*	Converts sound waves into an electrical signal.	A phone call, a voice assistant or a headset.

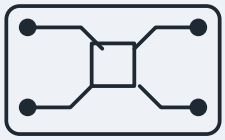
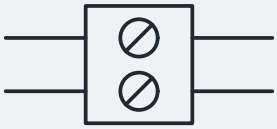
Actuator

Parts that turn electrical energy into something you can see, hear, feel or move: the loads and outputs.

Symbol	Name & appearance	Job	Real-world example
	**LED** *Looks like: A small coloured dome with two legs (the longer leg is positive), or a tiny surface-mount square.*	Emits light when current flows through it in the right direction.	A power indicator light, or a phone's screen backlight.
	Bulb *Looks like: A glass bulb with a thin filament inside, screwed onto a metal base.*	A load that converts electricity into light by heating a filament.	A room lamp, a torch or a car headlight.
	Motor *Looks like: A metal or plastic cylinder with a shaft sticking out and two terminals.*	A load that converts electricity into rotating motion.	A fan, a washing machine or an electric toothbrush.
	**Speaker / Buzzer** *Looks like: A round paper or plastic cone in a frame (speaker), or a small black cylinder with a hole on top (buzzer).*	A load that converts electricity into sound by vibrating air.	A phone loudspeaker, or the beep of a microwave oven.
	Heater *Looks like: A coil of resistive wire or a ceramic or metal element that glows orange when hot.*	A load that converts electricity into heat.	A toaster, an electric kettle or a 3D printer's hot end.
	Display (LCD/OLED) *Looks like: A flat glass panel with a ribbon cable or a 4-pin header. OLED pixels glow by themselves; LCDs need a backlight.*	Shows visual output such as text, numbers and pictures.	A smartwatch screen or a microwave's clock.

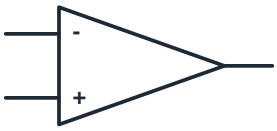
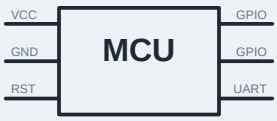
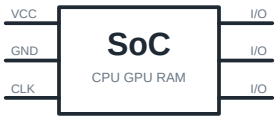
Connection

Parts that carry power and signals around, or join the circuit to the outside world.

Symbol	Name & appearance	Job	Real-world example
	Wire *Looks like: A copper conductor, solid or stranded, wrapped in coloured plastic insulation.*	Carries current between components.	Hook-up wire joining parts on a breadboard.
	**PCB (Printed Circuit Board)** *Looks like: A flat fibreglass board, usually green, with copper tracks, pads and holes. (No schematic symbol: board outline shown.)*	Holds components in place and connects them with etched copper tracks.	The motherboard in a computer, or the board in a TV remote.
	Connector *Looks like: A plug-and-socket pair such as USB, header pins or a barrel jack.*	Joins wires or cables so they can be connected and disconnected.	A USB cable plugging a phone into its charger.
	**Terminal Block** *Looks like: A green or blue plastic block with screw heads on top and wire holes on the side.*	Clamps bare wires under a screw for a firm, removable connection.	Wiring a power supply to a motor in a control panel.
	Antenna *Looks like: A metal rod (whip), a coiled wire, a flat PCB track or a metal dish.*	Transmits or receives radio waves.	A Wi-Fi router's antennas, or a telescopic FM radio aerial.

IC

Integrated circuits: thousands to billions of tiny components packed into a single chip.

Symbol	Name & appearance	Job	Real-world example
	**Op-Amp (IC)** *Looks like: A small 8-pin chip (DIP or surface-mount) with a notch marking pin 1, such as an LM358.*	Amplifies the difference between two inputs, or compares them and flips its output.	A microphone preamplifier, or a circuit that triggers when a voltage crosses a threshold.
	**Microcontroller (MCU)** *Looks like: A black rectangular chip (DIP or square QFP) such as the ATmega328P, often on a dev board like an Arduino.*	Runs a program to read inputs and control the circuit.	The brain of a washing machine, a remote control or an Arduino project.
	**SoC (System on Chip)** *Looks like: A square chip with hundreds of hidden contacts underneath, sometimes under a metal heat spreader.*	Puts a whole computer (CPU, GPU, memory and more) on one chip.	A smartphone processor, or the chip on a Raspberry Pi.
	ASIC *Looks like: A custom-marked black chip, often large, with a part number you will not find in standard catalogues.*	A chip designed from scratch to do one specific job extremely well.	Bitcoin mining chips, or a camera's image-processing chip.
	FPGA *Looks like: A square chip with many pins, usually found on a development board with a USB programmer.*	A chip full of logic blocks you can rewire in software to become different digital circuits.	Prototyping a new processor, or high-speed video processing.
	RAM *Looks like: Rectangular memory chips on a stick (DIMM), or small black chips soldered to a board.*	Holds data temporarily while the power is on; it is lost when power is removed.	The memory sticks in a computer, or the RAM in a phone.
	Flash Memory *Looks like: A tiny 8-pin chip (SOIC) or a card or stick (SD card, USB drive, SSD) holding several chips.*	Keeps data permanently, even with the power off.	An SD card holding photos, or the firmware chip in a router.

Quick Reference Cheat Sheet

Component	Job
AC Mains	Supplies 240V 60Hz alternating current
Transformer	Steps AC voltage up or down
Rectifier (Diode Bridge)	Converts AC into pulsating DC
Filter Capacitor	Smooths pulsating DC into steady DC
Voltage Regulator (LDO)	Holds a fixed steady output voltage
Buck Converter	Steps DC voltage down efficiently
Boost Converter	Steps DC voltage up efficiently
Buck-Boost Converter	Steps DC up or down as needed
Inverter	Converts DC back into AC
Switch	Opens or closes the circuit (on/off)
Push Button	Momentary on/off input
Relay	Electrically controlled switch
Optocoupler	Isolates two circuits using light
Transistor (BJT)	Switches or amplifies signals
MOSFET	High-power electronic switch
Thyristor (SCR)	Latching switch for high power
Fuse	Melts to break circuit if current is too high
Circuit Breaker	Trips and resets to protect from overcurrent
Varistor (MOV)	Absorbs voltage spikes (surge protection)
Zener Diode	Allows reverse current at a precise voltage
Resistor	Limits current and drops voltage
Potentiometer	Adjustable resistor (knob/tuning)
Capacitor	Stores charge, filters signals, blocks DC
Inductor (Coil)	Stores energy magnetically, resists current change
Diode	Lets current flow one direction only
Crystal Oscillator	Provides precise timing/clock signal
LDR (Photoresistor)	Resistance changes with light
Thermistor	Resistance changes with temperature
Accelerometer	Senses motion and tilt
Gyroscope	Senses rotation
Humidity Sensor	Senses moisture in air
Pressure Sensor	Senses pressure/force
Image Sensor (CMOS)	Turns light into digital pixels
Microphone	Converts sound into electrical signal
LED	Emits light when current flows
Bulb	Converts electricity into light (load)
Motor	Converts electricity into motion (load)
Speaker / Buzzer	Converts electricity into sound (load)
Heater	Converts electricity into heat (load)
Display (LCD/OLED)	Shows visual output
Wire	Carries current between components
PCB (Printed Circuit Board)	Holds and connects components
Connector	Joins wires or cables
Terminal Block	Screw connection for wires
Antenna	Transmits or receives radio waves
Op-Amp (IC)	Amplifies or compares signals
Microcontroller (MCU)	Runs a program to control the circuit
SoC (System on Chip)	Full computer (CPU+GPU+memory) on one chip
ASIC	Custom chip built for one specific task
FPGA	Reprogrammable logic chip
RAM	Temporary data storage
Flash Memory	Permanent data storage
Ground (0 V reference)	Gives a common 0 V reference point
Battery	Supplies DC from stored chemical energy